Series: Method of Differences

Questions 1–35

Question 1. Find the sum of the following series. You do not have to simplify your answer.

a)
$$1-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\frac{1}{3}-\frac{1}{4}+\cdots+\frac{1}{23}-\frac{1}{24}+\frac{1}{24}-\frac{1}{25}$$

b)
$$1-\frac{1}{4}+\frac{1}{4}-\frac{1}{7}+\frac{1}{7}-\frac{1}{10}+\cdots+\frac{1}{3n-5}-\frac{1}{3n-2}+\frac{1}{3n-2}-\frac{1}{3n+1}$$

c)
$$\frac{1}{6}-\frac{1}{10}+\frac{1}{8}-\frac{1}{12}+\frac{1}{10}-\frac{1}{14}+\frac{1}{12}-\frac{1}{16}\cdots$$
$$+\frac{1}{2(n+1)}-\frac{1}{2(n+3)}+\frac{1}{2(n+2)}-\frac{1}{2(n+4)}$$

d)
$$2-1+3-2+4-3+\cdots+n-(n-1)+(n+1)-n$$

e)
$$1\times 2-1\times 0+2\times 3-2\times 1+3\times 4-3\times 2+\cdots+n(n+1)-n(n-1)$$

Question 2. Write each of the following as a single fraction:

a) $\frac{1}{n+1}-\frac{1}{n+2}$

b) $1+\frac{1}{n+1}$

c) $\frac{1}{2}-\frac{1}{n+2}$

d) $-\frac{3}{2}+\frac{1}{n+2}+\frac{1}{n+1}$

e) $\frac{1}{12}+\frac{1}{2(n+3)}-\frac{1}{2(n+2)}$

Question 3. Use method of differences to find

$$\sum_{r=1}^{50}\left(\frac{1}{r+2}-\frac{1}{r}\right)$$

giving your answer in exact form.

Question 4. Use method of differences to find

$$\sum_{r=8}^{100}\left(\frac{1}{r+1}-\frac{1}{r}\right)$$

giving your answer in exact form.

Question 5. Use method of differences to find

$$\sum_{r=1}^{n}\left(\frac{1}{r+8}-\frac{1}{r+7}\right)$$

Question 6. Use method of differences to show that

$$\sum_{r=1}^{n}\left(\frac{1}{r+3}-\frac{1}{r+4}\right)=\frac{n}{4(n+4)}$$

Question 7. Use method of differences to find

$$\sum_{r=1}^{n}\left(\frac{1}{r+5}-\frac{1}{r+3}\right)$$

Question 8.

a) Show that

$$\frac{1}{r+4}-\frac{1}{r+5}=\frac{1}{(r+4)(r+5)}$$

b) Hence find

$$\sum_{r=1}^{n}\frac{1}{(r+4)(r+5)}$$

Question 9.

a) Write the following as a single fraction:

$$\frac{1}{r+1} - \frac{1}{r+2}$$

b) Hence show that

$$\sum_{r=1}^{n} \frac{1}{(r+1)(r+2)} = \frac{n}{an+b}$$

where a and b are constants to be found.

Question 10.

a) Show that

$$\frac{1}{r-1} - \frac{1}{r+1} = \frac{2}{(r-1)(r+1)}$$

b) Hence find

$$\sum_{r=5}^{50} \frac{1}{r^2-1}$$

giving your answer in exact form.

Question 11.

a) Show that

$$\frac{1}{r+3} - \frac{1}{r+4} = \frac{1}{r^2+7r+12}$$

b) Hence find

$$\sum_{r=8}^{22} \frac{1}{r^2+7r+12}$$

giving your answer in exact form.

Question 12.

a) Show that

$$\frac{1}{r-1} - \frac{1}{r+2} = \frac{3}{(r-1)(r+2)}$$

b) Hence find

$$\sum_{r=2}^{n} \frac{1}{(r-1)(r+2)}$$

giving your answer in exact form.

Question 13. By considering

$$\frac{1}{r^3} - \frac{1}{(r+1)^3}$$

find

$$\sum_{r=1}^{n} \frac{3r^2 + 3r + 1}{r^3(r+1)^3}$$

Question 14. By considering

$$\frac{1}{r-2} - \frac{1}{r}$$

find

$$\sum_{r=3}^{n} \frac{1}{r^2 - 2r}$$

as a single fraction.

Question 15.

a) Find

$$\frac{1}{2r+1} - \frac{1}{2r+3}$$

as a single fraction.

b) Find the sum of the series

$$\frac{2}{3\times 5} + \frac{2}{5\times 7} + \frac{2}{7\times 9} + \cdots + \frac{2}{31\times 33}$$

giving your answer in exact form.

Question 16.

a) Write

$$\frac{1}{r} - \frac{1}{r+1}$$

as a single fraction.

b) Find how many terms of the series

$$\frac{1}{1\times 2} + \frac{1}{2\times 3} + \frac{1}{3\times 4} + \cdots + \frac{1}{r(r+1)} + \cdots$$

are needed to exceed 0.98

Question 17.

a) Write

$$\frac{1}{2r} - \frac{1}{2r+2}$$

as a single fraction.

b) Find

$$\sum_{r=1}^{n} \frac{1}{2r(r+1)}$$

c) Hence find

$$\sum_{r=1}^{\infty} \frac{1}{2r(r+1)}$$

Question 18.

a) Show that

$$r+1 = \frac{1}{2}[(r+1)(r+2) - r(r+1)]$$

b) Hence use method of differences to find

$$\sum_{r=1}^{n} 2(r+1)$$

Question 19.

a) Show that

$$(r+1)(r+2) \equiv \frac{1}{3}[(r+1)(r+2)(r+3) - r(r+1)(r+2)]$$

b) Hence use method of differences to find

$$\sum_{r=1}^{n} 3(r+1)(r+2)$$

Question 20. Find the smallest integer n such that

$$\sum_{r=1}^{n} \left(\frac{2}{r^2} - \frac{2}{(r+1)^2} \right) > 1.95$$

Question 21.

a) Show that

$$\frac{1}{r!} - \frac{1}{(r+1)!} = \frac{r}{(r+1)r!}$$

b) Hence show that

$$\sum_{r=1}^{n} \frac{r}{(r+1)r!} = a - \frac{b}{(n+1)!}$$

where a and b are constants to be found.

Question 22.

a) Expand and simplify $(r+2)^3 - r^3$

b) Hence find

$$\sum_{r=1}^{n} 3r^2 + 6r + 4$$

giving your answer in fully factorised form.

c) By considering your answer to part b), find

$$\sum_{r=1}^{n} r(r+2)$$

Question 23.

a) Show that

$$(2r+1)^3 - (2r-1)^3 = 24r^2 + 2$$

b) Hence, using method of differences, show that

$$\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$$

Question 24.

a) Show that

$$(2r+1)^4 - (2r-1)^4 = 64r^3 + 16r$$

b) Hence, using method of differences and the summation series formulae, show that

$$\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$$

Question 25.

a) Show that

$$(2r+1)^5 - (2r-1)^5 = 160r^4 + 80r^2 + 2$$

b) Hence, using method of differences and the summation series formulae, show that

$$\sum_{r=1}^{n} r^4 = \frac{1}{30}n(n+1)(2n+1)(3n^2+3n-1)$$

Question 26. Find the value of k such that

$$\sum_{r=k}^{100}\left(\frac{2}{r} - \frac{2}{r+1}\right) = \frac{65}{1818}$$

Question 27.

a) Show that

$$\frac{2}{r} - \frac{1}{r+1} - \frac{1}{r+2} = \frac{3r+4}{r(r+1)(r+2)}$$

b) Find

$$\sum_{r=1}^{n} \frac{3r+4}{r(r+1)(r+2)}$$

c) Find the value of k such that

$$\sum_{r=k}^{25} \frac{3r+4}{r(r+1)(r+2)} = \frac{1174}{12\,285}$$

Question 28.

a) Show that

$$2^{r+1}(r-1) - 2^r(r-2) = r2^r$$

b) Hence show that

$$\sum_{r=1}^{n} r2^r = 2 + 2^{n+1}(n-1)$$

Question 29.

a) Show that

$$\ln(r+2) - \ln(r+1) = \ln\left(1 + \frac{1}{r+1}\right)$$

b) Hence show that

$$\sum_{r=1}^{n} \ln\left(1 + \frac{1}{r+1}\right) = \ln\left(\frac{n+a}{b}\right)$$

where a and b are integers to be found.

Question 30. Use method of differences to write

$$\sum_{r=1}^{n} \ln\left(\frac{r+1}{r}\right)$$

as a single function.

Question 31. Find the exact value of

$$\sum_{r=21}^{50} \ln\left(\frac{r+1}{r-1}\right)$$

Question 32.

a) Show that

$$\sum_{r=1}^{n} \ln\left(\frac{r+3}{r+1}\right) = \ln(a(n+b)(n+c))$$

where a, b and c are positive constants to be found.

b) Hence find

$$\sum_{r=1}^{n} \ln\left(\frac{r+3}{r+1}\right)^{36}$$

Question 33.

a) Show that

$$\sum_{r=1}^{n} \ln\left(\frac{2r+3}{2r+1}\right) = \ln(a(bn+c))$$

where a, b and c are positive constants to be found.

b) Hence find the smallest value of n for which

$$\sum_{r=1}^{n} \ln\left(\frac{2r+3}{2r+1}\right) > 4$$

Question 34. Find

$$\sum_{r=6}^{\infty} \left(\frac{1}{r+1} - \frac{1}{r+3}\right)$$

Question 35.

a) Show that

$$\frac{1}{2(r+2)} - \frac{1}{2(r+4)} = \frac{1}{(r+2)(r+4)}$$

b) Show that if

$$\frac{7}{24} - \sum_{r=1}^{n} \frac{1}{(r+2)(r+4)} < \frac{19}{180}$$

then

$$n \geq 7$$